

System Requirement Specification (SRS)

MERN Stack Retail Inventory, Sales & Debt Management System

Target Region: Somalia Retail Sector

Architecture: Offline-First MERN

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1. PROJECT OVERVIEW

The objective of this project is to develop a modern, high-performance web application designed for retail shops in Somalia. The system replaces vulnerable paper ledger management with an automated solution for real-time inventory tracking, low-stock alerts, customer debt tracking with dynamic audit trails, and multi-channel messaging support (SMS & WhatsApp). It operates with an **Offline-First** strategy for uninterrupted point-of-sale (POS) operations during internet disruptions.

2. CORE ARCHITECTURE & TECH STACK

Component	Technology & Description
Frontend	React.js, Redux Toolkit, Tailwind CSS, Lucide Icons
Backend	Node.js, Express.js (RESTful API Services)
Database	MongoDB with Mongoose Object Data Modeling (ODM)
Offline Storage & Sync	Browser IndexedDB & Service Workers for Local Queue Management
Messaging Gateway	Twilio SMS API & WhatsApp Business API Integration
Hardware Peripherals	Web Print API integration for 80mm & 58mm Thermal Printers

3. FUNCTIONAL REQUIREMENTS

A. Point of Sale (POS) Module (Offline-First)

- Fast Product Lookup:** Instant search filtering by product name, category, or automated barcode scanner input.
- Offline Transaction Engine:** Seamless ability to process sales during internet outages. Data is stored locally in IndexedDB and automatically synchronized to MongoDB upon reconnection.
- Payment Gateway Options:** Built-in support for Cash, Mobile Money (EVC Plus, Zaad, Sahal, eDahab), and Credited Sales (Debt).
- Thermal Receipt Printing:** Instant layout rendering formatted for 80mm/58mm paper size containing store info, line items, totals, and remaining customer debt balance.

B. Inventory & Stock Management

- Dual-Pricing Model:** Tracks both *Cost Price (Purchase Price)* and *Selling Price* per product for accurate margin analytics.
- Low-Stock Automated Alerts:** Visual indicator flags (Red alerts) on the dashboard when stock quantities fall below specified threshold limits (e.g., ≤ 5 units).

- **Expiry Date Monitoring:** Tracks perishable goods (beverages, food products, cosmetics) with expiration warnings to minimize product waste.

C. Somali Customer Debt Ledger (Buugga Deynta)

- **Customer Accounts:** Manage customer records linked by full name and verified phone number.
- **Partial & Full Repayments:** Enables cashiers or admins to record incremental balance payments with automatic credit deduction.
- **Audit Trail Timeline:** Chronological record detailing every debt issuance and partial payment with exact timestamps, previous balance, paid amount, and updated balance.
- **Automated Notifications (SMS & WhatsApp):**
 - *Credit Sale Alert:* Instant dispatch specifying purchase date, item summary, amount borrowed, and total accumulated debt.
 - *Payment Confirmation:* Direct electronic receipt confirming payment received and remaining unpaid balance.

D. Business Analytics & Financial Insights

- **Movement Rate Analytics:** Identifies high-velocity items (*Fast-Moving*) vs. stagnant inventory (*Dead Stock*).
- **Net Profit Calculation Formula:**

$$Net\ Profit = \sum (Selling\ Price - Cost\ Price) \times Quantity\ Sold$$

4. USER ROLES & ACCESS CONTROL (RBAC)

System Feature / Operation	Admin (Owner)	Cashier (Staff)
POS Checkout & Transaction Processing	Yes	Yes
Record Debt & Process Partial Repayments	Yes	Yes
Print Receipts (Thermal Printer)	Yes	Yes
View Product Cost Price & Net Profit Analytics	Yes	No
Add/Edit Products & Thresholds	Yes	No
User Management (Create/Remove Accounts)	Yes	No

5. DATABASE BLUEPRINT STRUCTURE

```
[Users] (Role: Admin / Cashier)
├── [Products] (Barcode, Name, CostPrice, SellingPrice, Quantity, MinAlert, ExpiryDate)
├── [Customers] (Name, Phone, TotalDebt)
│   └── [DebtHistory] (Type: DEBT_ADDED / PAYMENT_MADE, Amount, RemainingBalance, Date)
└── [Sales] (Items, SubTotal, PaymentType, CustomerRef, SyncStatus)
```

6. NON-FUNCTIONAL REQUIREMENTS

- **Performance:** POS barcode scanning and lookup response time under 100 ms.
- **Data Integrity & Reliability:** Conflict-free synchronization algorithm ensuring no duplicate or dropped transactions during state change from offline to online.
- **Usability:** Touch-friendly, streamlined user interface requiring minimal clicks for high-volume transactions.